

Immunoblotting

Risk assessment

- ▷ Wear gloves, safety glasses, lab coat
- Dispose milk in accordance with local regulations
- Some jurisdictions prohibit draining dairy products due to their biological oxygen demand



Reviewed: Jan 15, 2023

This is a bench card. Full protocol available online.

>> Immunoprobng

- | | |
|---|---|
| □ Incubation tray or Resealable plastic bag | □ Bovine serum albumin or Nonfat dry milk powder ("Blotto") |
| □ R0169 TBS with Tween™ 20, 1 L | □ Antibodies, primary and secondary |

- (1.) Prepare blocking buffer by diluting Blotto to 5% or BSA to 1–5% in 20 mL TBST.

Critical: Avoid milk-based blocking buffers such as Blotto for phospho-proteins or streptavidin-based detection. Milk contains phosphatases and biotin, respectively. Phosphatase activity can be quenched with 50 mM sodium fluoride.

- (2.) Block the membrane face-up in 5 mL blocking buffer for 30 min at room temperature with agitation.

- (3.) Dilute the primary antibody in blocking buffer (typically 1:1 000 to 1:10 000).

Critical: For multiplexed experiments, avoid combining mouse and rat primary antibodies as they are difficult to cross-adsorb. Chicken antibodies tend to non-specifically bind to PVDF and nylon-based membranes, leading to high background.

- (4.) Discard blocking buffer. Incubate with diluted primary antibody for 30–60 min at room temperature or overnight at 4 °C with agitation.

- (5.) Transfer membrane to a clean tray.

- (6.) Wash four times with 10–20 mL TBST for 3 min each with gentle shaking.

- (7.) Dilute secondary antibody in TBST (typically 1:2 000 to 1:10 000) according to vendor instructions.

Critical: Beware, secondary anti-mouse antibodies may be subclass-specific for IgG1, IgG2a, or IgG2b. Use highly cross-adsorbed secondaries when multiplexing with antibodies from different species.

- (8.) Incubate with secondary antibody at room temperature for 30–60 min with agitation.

- (9.) Wash again as before.

A > Fluorescent detection

- (1.) Specifications for infrared dyes used, for example, with LI-COR Odyssey® Imagers:

- Use IRDye® 800CW or IRDye® 680RD secondary antibodies, at 1:20 000 dilution.
- Use IRDye® 800CW for the lowest-abundance target; it gives the lowest background.
- Rinse the membrane with 1 × TBS to remove residual Tween™ 20.

- (2.) Clean the imager surface with distilled water, then 70% ethanol. Dry with a lint-free wipe.

- (3.) Scan the membrane using a suitable fluorescence imager.

B > **Luminescent detection**

- | | |
|---|---|
| ☐  R0118 1 M Tris hydrochloride, 100 mL | ☐  R0120 200 × <i>p</i> -Coumaric acid, 10 mL |
| ☐  R0119 100 × Luminol, 10 mL | ☐  R0121 ECL reagent, 10 mL |

- (1.) Prepare 10 mL ECL reagent per membrane just before use.
- (2.) Drip-dry the membrane and distribute ECL reagent dropwise to cover the surface evenly. Incubate for  1 min at room temperature.
- (3.) Briefly slip the membrane through deionized water and gently drag over the edge of a container to remove excess liquid.
- (4.) Expose to film or camera. The signal will last for at least 10 min. 

Stripping and reusing membranes

- | |
|--|
| ☐  R0122 Stripping buffer, 15 mL |
|--|

- (1.) Wash membrane in distilled water for 5 min.
- (2.) Strip antibodies using one of the following:

Stripping buffer	NC	PVDF
0.2 M NaOH		
25 mM Glycine, pH 2.0		
Stripping buffer, pH 6.8	Not recommended	

Critical: Do not use high concentration of SDS with nitrocellulose membranes as this can destroy membrane integrity. 

- (3.) Wash membrane twice in distilled water for 5 min.
- (4.) Reblock and reprobe as needed. 

 Recipe (available online)  Troubleshooting (available online)  Notes (available online)

Lab Protocols — © B. C. Buchmuller. Licensed CC-BY-SA-4.0. Research use; provided as-is, without warranty. Current when printed. Visit <https://benjbuch.github.io/check/?q=e1dd568> or scan the QR code to read the full protocol or to check for updates.

